# Emergency Department Wait Time Prediction Using Machine Learning

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***Abstract*—Long, unpredictable waits in the Emergency Department (ED) damage patient outcomes, raise clinical risk, and degrade hospital-wide flow, yet the triage tools used today rank urgency without telling anyone how long care will actually take. We address this transparency gap by building and comparing five regression models—Linear Regression, Ridge, Random Forest, XGBoost, and LightGBM—that estimate ED length of stay (LOS) in minutes from triage features alone, using a publicly available CC0 dataset of 9,216 ED visits expanded from 17 raw fields to 26 engineered features. Beyond the standard overall-accuracy report, we audit the model along three axes that the literature usually treats in isolation: per-Emergency-Severity-Index (ESI) performance across all five acuity levels, SHAP-based feature attribution, and fairness across race/ethnicity, gender, and age group. On a held-out 20% test set the regularized linear baseline gave the best aggregate error (MAE 89.1 min, $R^2$ 0.141), narrowly ahead of Random Forest (MAE 91.0) and ahead of XGBoost (93.8) and LightGBM (93.4); SHAP confirmed that the ESI acuity score alone carries roughly six times the explanatory weight of any other feature; ESI-stratified evaluation revealed a near fourfold MAE gap between Level 5 (37 min) and Level 2 (130 min) patients that the pooled metric hides; and the fairness audit surfaced a 15.3-minute MAE gap across race/ethnicity groups and an 11.5-minute gap across age groups, with only a 1.0-minute gap between genders. The work is intended for ED operations leads who need defensible patient-flow estimates, clinicians who must communicate realistic timing at triage, equity researchers auditing ML deployments in healthcare, and patients who deserve a transparent answer to "how long will I wait?".**

***Index Terms*—emergency department, wait time prediction, length of stay, machine learning, regression, triage, ESI, XGBoost, LightGBM, SHAP, algorithmic fairness**

## I. Introduction

### A. Background and Motivation

The Emergency Department (ED) is the open door of the modern hospital: it is where unscheduled, time-sensitive, and clinically heterogeneous patients all arrive on the same shift and compete for the same finite pool of beds, nurses, and physicians. Across the United States, Europe, and much of the Asia-Pacific region, ED visit volumes have outpaced hospital capacity for more than a decade, and the resulting overcrowding is now classified as a genuine public-health risk rather than an operational nuisance [15], [23]. The downstream effects are well documented. Patients who wait too long are more likely to suffer adverse events while they wait, more likely to leave without being seen, and statistically more likely to die in hospital after admission [24]. From the staff side, sustained overcrowding drives burnout, increases handoff errors, and pushes ambulance diversion, which spreads the bottleneck to the surrounding emergency medical system.

A second motivating thread is operational. Length of stay in the ED has become one of the most-watched performance indicators in emergency medicine because it is a single number that condenses everything happening to a patient between triage and disposition: how long the workup took, whether the consult arrived, whether an inpatient bed was available, whether the lab was busy [5]. When ED LOS becomes unpredictable, a chain of hospital-wide planning processes degrades with it—bed assignment, transport scheduling, nurse staffing, even meal delivery. A predictive model that turns LOS from a guess into a number is therefore valuable not only at the bedside but at the operations desk.

The third thread is the gap between urgency and timing. The triage scoring systems EDs already use—primarily the Emergency Severity Index (ESI) in the United States and the Manchester Triage Scale (MTS) in Europe [16]—answer the question "who do we see next?" but not "how long until I am seen?" or "how long until I go home?". These two questions are clinically and operationally distinct. Machine Learning (ML) models trained on the same routinely collected triage data—vital signs, demographics, arrival mode, ESI level—can fill the second question, producing a continuous patient-level estimate that augments rather than replaces the existing triage scale. A growing body of recent work [3], [7], [9] has begun to populate this space, but most published efforts still cast the problem as a binary "prolonged vs. not prolonged" classification, train on proprietary single-center data, or report a single pooled accuracy metric that hides performance differences between patient subgroups.

### B. Research Question

The problem we address is the absence, in current practice, of a transparent patient-level estimate of how long an ED visit will take. The estimate must be computable at the moment of triage, must use only data already collected during the standard triage interaction, must come with an explanation that clinical staff can interpret, and must behave consistently across patient subgroups—a model that is twice as wrong for one demographic group as for another is not deployable in a clinical setting regardless of its average accuracy. We split the

problem into two research questions, each chosen to target a different facet of trustworthiness:

- **RQ1 (Predictive accuracy and generalizability across triage levels).** Can supervised regression models trained on routinely collected ED triage data predict patient LOS in minutes with materially lower error than naive baselines, and does that accuracy generalize across all five ESI triage levels, or is it driven by a single dominant sub-population?
- **RQ2 (Explainability, interpretability, and demographic fairness).** Which features actually drive the model's LOS predictions according to SHAP attribution, do those drivers align with clinical intuition and with the patterns reported in prior interpretable-ML studies, and does the resulting model produce equitable error across subgroups defined by race/ethnicity, gender, and age?

### C. Objectives

The research questions translate to four concrete project objectives:

1) Train and benchmark five regression models—Linear Regression, Ridge, Random Forest, XGBoost, and LightGBM—on an 80/20 hold-out split with 5-fold cross-validation on the training fold, reporting MAE, RMSE, and $R^2$.
2) Refit the best model separately within each of the five ESI levels and compare per-level error, exposing the subgroup variation that pooled metrics hide.
3) Apply SHAP TreeExplainer to the best gradient-boosted model and report a global feature ranking, both for clinical sanity-checking and for direct comparison with previously published SHAP rankings.
4) Disaggregate the test-set MAE and RMSE across race/ethnicity, gender, and age group, and quantify the worst-case subgroup gap.

### D. Contributions

Against the existing literature, this paper offers four contributions:

1) **A combined methodology.** Our pipeline simultaneously delivers continuous LOS regression, ESI-stratified evaluation, SHAP interpretability, and demographic fairness auditing on a single publicly accessible dataset. None of the twelve studies surveyed in Section II delivers all four together.
2) **An empirical counter-finding.** On a moderately sized dataset, regularized linear regression matches or slightly beats the gradient-boosted models, in tension with the prevailing claim that XGBoost and LightGBM dominate ED prediction across the board [5], [9]. We argue this is a dataset-size effect worth reporting honestly rather than tuning around.
3) **An equity finding.** A non-trivial 15.3-minute MAE gap across race/ethnicity groups exists in the model's errors and is invisible if only a single pooled metric is reported. This argues for subgroup auditing as a default, not an optional extension, in ED ML.
4) **A reproducibility artifact.** A cleaned 26-feature dataset, a documented analysis notebook, and saved per-model and per-subgroup metrics are released alongside this paper so that other teams can replicate, criticize, or extend the experiment.

### E. Paper Roadmap

Section II reviews twelve recent peer-reviewed studies on ML for ED triage and patient-flow prediction, ending with an explicit list of recurring gaps and a positioning paragraph that locates our contribution inside the literature. Section III covers the dataset, a condensed preprocessing pipeline, the five candidate models, and the evaluation protocol. Section IV reports results in four layers—overall, ESI-stratified, SHAP-based, and fairness-disaggregated— and compares each layer against the closest prior-work benchmark. Section V interprets the results, lists strengths and limitations honestly, and pulls out practical implications. Section VI concludes and proposes future work. All numerical tables are collected in the Appendix and referenced inline by number throughout.

## II. Literature Review

### A. Theoretical Background

Three research traditions converge on the ED prediction problem. The first is clinical triage, which since the late 1990s has been formalized through ordinal scales such as ESI [16] that translate vital signs and chief complaint into a 1–5 priority. The second is the rise of gradient-boosted decision trees as the default tabular learner in healthcare [17]–[19]; on most clinical structured-data problems they sit at the Pareto front of accuracy and training cost, which is why they dominate the recent ED literature. The third is post-hoc explainability through SHAP [14], [20], which lets a tree ensemble be interrogated at the feature level without sacrificing predictive accuracy. The standard recipe in the recent ED literature is to combine all three: feed triage features into a boosted ensemble, evaluate on AUC for classification or on MAE/RMSE/$R^2$ for regression [9], [10], and explain the result with SHAP. Our study follows this recipe and tests where it breaks.

### B. Synthesis of the Twelve Reviewed Studies

We organize the twelve studies into five themes: systematic reviews, empirical wait-time and LOS models, benchmark and interpretability work, fairness and overcrowding studies, and advanced AI paradigms. The full synthesis appears as Table I in Appendix A; the reading of the table that follows is the narrative the table is designed to support.

### C. Systematic Reviews

Two systematic reviews anchor the field. Porto [1] surveys ML and natural-language-processing approaches to ED triage in a 2024 PRISMA-2020 review and concludes that learned models consistently outperform rule-based triage scales, while flagging the absence of continuous-wait-time regression and

the limited use of explainability tools as the two largest unmet needs. The second 2025 systematic review [2] surveys ML for predicting inpatient admission from ED triage and reports a tight AUC range of 0.81–0.93 across rigorous studies, with random forest and gradient boosting at the top of the leaderboard.

### D. Empirical Wait-Time and LOS Models

Five empirical studies form the most directly comparable group to our work. Wang et al. [3] train an interpretable XGBoost classifier on prolonged ($\geq$30 min) wait time in ESI-3 patients, achieving AUC $\approx$ 0.72 and documenting performance gaps across race/ethnicity and insurance type, but remain restricted to the binary target and a single ESI bucket. Ricciardi et al. [5] apply random forest, decision tree, and SVM to nearly half a million Italian ED admissions and reach AUC $\approx$ 0.80 for the binary prolonged-LOS target, validating that triage features alone carry useful signal. Seo et al. [6] extend the feature space with NLP-processed chief-complaint text and find LightGBM the best of their suite. Wang, Sambamoorthi et al. [7] compare five learners on the same prolonged-wait-time target as [3] and confirm XGBoost/RF as the strongest. Wu et al. [9] stand out as the clearest precedent for our setup: they treat ED LOS as a continuous regression target on 187,028 records, identify LightGBM as the strongest model, and surface "current emergency waiting time"—a real-time queue signal—as the dominant feature.

### E. Benchmarks and Interpretability

Two studies build infrastructure rather than fit models on their own. Xie et al. [4] release the MIMIC-IV-ED benchmark and a reference preprocessing pipeline for the wider research community, demonstrating that gradient boosting and random forest beat traditional clinical scoring systems. Sulaiman et al. [8] push the explainability frontier past SHAP by combining ensemble ML with TE2Rules rule extraction on ten minimal features, sketching a path toward clinician-readable "if-then" explanations.

### F. Fairness, Overcrowding, and Advanced AI

The remaining three studies cover the wider context. Vural et al. [10] target system-level overcrowding rather than individual-patient prediction, using time-series transformers (TSiTPlus, XCMPlus) on 6-hour and 24-hour occupancy and reporting MAE $\approx$ 4.19 patients with $R^2 \approx 0.56$. Lett et al. [11] compare intersectional and marginal fairness debiasing for ED admission prediction on MIMIC-IV-ED; both methods preserve overall AUROC at 0.85, but the intersectional approach roughly doubles the calibration improvement (22.3% vs. 11.3% subgroup error reduction). Sun et al. [12] introduce ED-Copilot, which pairs BioGPT with a reinforcement-learning agent to recommend lab orderings that minimize wait time and demonstrate a roughly 50% reduction in average laboratory wait time—the most ambitious AI paradigm in the review.

### G. Identified Research Gaps

Reading Table I together with the per-study accounts above makes five gaps visible. *Gap 1: continuous regression is rare*—ten of the twelve studies treat their target as a binary class; only Wu et al. [9] and Seo et al. [6] model wait time or LOS as a continuous variable in minutes. *Gap 2: triage segmentation is absent*—no study reports performance broken down across all five ESI levels; subgroup analyses, when they exist, focus on ESI-3 alone. *Gap 3: explainability is uneven*—only four of twelve studies apply SHAP, only one [8] explores rule-based alternatives, and seven publish no interpretability analysis at all. *Gap 4: reproducibility is limited*—nine of twelve studies use proprietary single-center data, which prevents direct replication. *Gap 5: fairness audits are scarce*—only two studies [3], [11] formally audit subgroup performance, and both restrict their audit to classification rather than continuous regression.

### H. Positioning of This Study

Our work is designed to attack all five gaps at once on a single publicly-released dataset. We model LOS as a continuous variable in minutes (addresses Gap 1), evaluate every model across all five ESI levels rather than collapsing to a binary cut (Gap 2), apply SHAP TreeExplainer to our best gradient-boosted model and publish the global ranking (Gap 3), use the CC0 Hospital Emergency Room Visits dataset and release a cleaned 26-feature derivative (Gap 4), and disaggregate test-set MAE and RMSE across race/ethnicity, gender, and age group (Gap 5). To our knowledge no prior ED study delivers all five.

## III. Methodology

### A. Dataset

The data come from the Hospital Emergency Room Visits dataset on Kaggle, released under a CC0 (public-domain) license, with a feature schema designed to mirror the openly published MIMIC-IV-ED benchmark [4], [13]. The raw file contains 9,216 ED visits and 17 fields covering structured triage data, vitals, demographics, arrival context, and disposition. Table II (Appendix) summarizes every raw field grouped by clinical role: identifier, demographics, arrival context, triage acuity, vital signs, and disposition. The target field is `ed_los_minutes`, the total minutes a patient spends in the department from arrival to physical departure. As Table III (Appendix) shows, the empirical distribution of this variable is heavily right-skewed—a mean of 189.8 min sitting well above the median of 153.7 min, and a long tail reaching 2,644.9 min for boarding cases—which is typical of ED stay-time data and motivates a non-Gaussian modeling treatment.

### B. Data Cleaning and Preprocessing

Preprocessing was applied as a single sequential pipeline. Numeric vital signs with up to 12% missingness were imputed by their median; the four categorical fields with sparse missing entries were imputed by their mode; clinical reference bounds (HR 30–250 bpm, SBP 60–280 mmHg, LOS 5–4,320 min, etc.) were used to confirm no implausible values survived

imputation; categorical features were label-encoded; five new features were engineered from existing ones (*shock_index* = HR/SBP, *pulse_pressure* = SBP−DBP, three-bucket *arrival_shift*, binary *is_weekend*, four-bucket *age_group*); and finally all 19 numeric model features were standardized to zero mean and unit variance, in line with the convention adopted by Wu et al. [9] and Wang et al. [3]. Table V (Appendix) gives the row-by-row before/after summary, ending with 9,216 records, 26 features, and zero remaining missing cells; Table IV profiles the missingness as it appeared on entry to the pipeline.

### C. Model Feature Set

After preprocessing, the regression input vector consists of 19 standardized features grouped by source: *patient demographics* (age, gender_encoded, race_encoded, age_group_encoded), *vital signs* (heartrate, sbp, dbp, temperature, o2sat, resprate), *triage information* (acuity, pain_score), *engineered physiology* (shock_index, pulse_pressure), *arrival context* (arrival_hour, is_weekend, arrival_transport_encoded, arrival_shift_encoded), and *visit-level disposition* (was_admitted). Pearson screening confirmed that only `acuity` has a non-trivial linear correlation with the target ($r = -0.38$); every other feature falls in $|r| \leq 0.02$, which is the empirical signature of a problem dominated by one strong driver plus weak nonlinear and interaction signal in the remainder.

### D. Models and Why We Chose Them

We evaluated five regressors covering three families. *Linear baselines*: Linear Regression and Ridge ($\alpha = 1.0$); these establish a reference floor and check whether L2-regularization helps under the multicollinearity present between SBP, DBP, and pulse_pressure. *Tree baseline*: Random Forest with 200 trees, max depth 14, and minimum 4 samples per leaf, following the strong RF results reported by [5] and the bagging argument of [17]. *Gradient boosting*: XGBoost [18] with 400 trees, learning rate 0.05, max depth 6, and 0.85 subsample/colsample fractions; and LightGBM [19] with 500 trees, 31 leaves, learning rate 0.05. The boosting hyperparameters were chosen by light manual tuning on the training fold to match the configurations most commonly reported in the ED ML literature [3], [6], [9] so that any difference in our findings cannot be charged to underpowered baselines. All models use `random_state=42`.

### E. Evaluation Protocol

The 9,216 records were split 80/20 into a training set (7,372 records) and a held-out test set (1,844 records) using a fixed seed. We report three metrics: MAE (interpretable in clinical units—minutes of error), RMSE (which over-weights large errors and so penalizes the long-tail boarding cases), and $R^2$ (proportion of LOS variance explained). All three are reported both on the held-out test set and as a 5-fold cross-validation average on the training fold. For ESI-stratified evaluation, we re-split the data within each acuity level before refitting Ridge to keep training- and test-fold separation. For SHAP analysis we apply TreeExplainer [14], [20] to the LightGBM model and report mean |SHAP| per feature on the test set. For the fairness audit, we compute MAE and RMSE separately for each group of the race/ethnicity, gender, and age-group categorical variables on the same held-out test set. The full software stack is Python 3.11 with scikit-learn 1.8.0, XGBoost 3.2.0, LightGBM 4.6.0, and SHAP 0.51.0; results files and the analysis notebook are released with the paper.

## IV. Results

### A. Overall Model Performance

Table VI (Appendix) reports test-set and 5-fold CV results for all five models in a single synthesis. Ridge Regression yielded the lowest test MAE at 89.09 min and the highest test $R^2$ at 0.141, edging out plain Linear Regression by less than 0.01 on every metric (the two collapse to nearly the same model when the regularization weight is small relative to the gradient of the loss surface). Random Forest came next at MAE 91.03; XGBoost and LightGBM trailed at 93.80 and 93.42. Cross-validation metrics tracked the held-out test ranking to within $\pm 1$ MAE point for every model, ruling out overfitting as an explanation. As a sanity floor, the naive "predict the training-set mean" baseline produced a test MAE of 98.71, so the weakest of our five models still recovers around 5 minutes of error reduction over guessing the average.

*Comparison with prior work.* Wu et al. [9], the most directly comparable continuous-regression study, achieve LightGBM $R^2$ between roughly 0.40 and 0.55 on 187,028 records, with the bulk of their predictive lift coming from a real-time "current emergency waiting time" queue-depth feature that is unavailable in our static dataset. Ricciardi et al. [5] report AUC $\approx$ 0.80 on a binary prolonged-LOS target on roughly 500,000 records—a different metric on a different framing, but a useful upper anchor for what triage features alone can deliver. The most striking finding in our table, given the literature backdrop, is that on this dataset *regularized linear regression matches the best non-linear model*. The straightforward explanation, consistent with the correlation profile of the features ($r = -0.38$ for acuity, $|r| \leq 0.02$ for everything else), is that this prediction problem contains one strong linear driver and very little exploitable interaction signal at the available sample size. Boosting needs interactions to pay for its capacity; on a dataset of 9,216 records with one dominant linear effect, regularized linear models can match it.

### B. ESI-Stratified Performance

Refitting Ridge separately within each acuity level produced Table VII (Appendix). Two patterns emerge that the pooled metric hides. First, MAE scales almost linearly with mean LOS by triage level: ESI-5 patients (mean LOS 91 min, lowest acuity) have MAE 36.96 min, while ESI-2 patients (mean LOS 265 min, second-highest acuity) have MAE 130.28 min—a 3.5$\times$ ratio that the single pooled MAE of 89 min would never reveal. Second, every per-level $R^2$ value sits between $-0.16$ and $-0.01$, even though the pooled $R^2$ is 0.141. This is not a contradiction. The pooled $R^2$ benefits from

the strong between-level variance driven by acuity itself; once that variance is removed by stratifying, the residual within-level signal in vital signs and demographics is too weak for the linear model to capture. The clinical implication is that a pooled-metric deployment would systematically over-promise accuracy for high-acuity patients and under-promise it for low-acuity patients. We did not find this pattern reported in any of the twelve reviewed studies.

### C. SHAP Feature Importance

Table VIII (Appendix) reports global mean |SHAP| values for the LightGBM model on the held-out test set, ranked from largest to smallest. The ESI `acuity` score dominates at 46.07—approximately 6.4× the second feature. The next tier consists of physiology: `o2sat` (7.15), `dbp` (6.27), `temperature` (5.88), `age` (5.87), `pulse_pressure` (5.85), `resprate` (5.20), `shock_index` (5.07), `heartrate` (4.96), `sbp` (3.94). Demographics, arrival context, and visit-level disposition—features that a fairness-skeptical reader might worry the model leans on—occupy the bottom half of the ranking, with `age_group_encoded` dead last at 0.62.

*Comparison with prior work.* The dominance of the acuity score matches Wang et al. [3], whose XGBoost SHAP analysis put ESI level at the top of the list, and Wang, Sambamoorthi et al. [7], who placed ESI level and arrival hour as their top two. Our `arrival_hour` ranks 13th rather than near the top, which we attribute to two differences: the prior studies' EDs had strong queue-depth dynamics that load onto arrival hour as a proxy, and our dataset has a more uniform arrival-hour distribution that flattens this proxy effect. The high SHAP rank of vital signs over administrative and demographic features matches the clinically intuitive story that physiological severity drives LOS more than identity does.

### D. Demographic Fairness Audit

Tables IX, X, and XI (all in Appendix) report MAE and RMSE for the Ridge model disaggregated across race/ethnicity, gender, and age group, respectively. The headline numbers are a 15.26-minute MAE gap between the worst-served group (Black/African American, MAE 91.23) and the best-served group (OTHER, MAE 75.97); a 1.03-minute MAE gap between female (88.61) and male (89.64) patients; and an 11.50-minute MAE gap between the worst-served age group (Pediatric, MAE 95.60) and the best-served (Middle Age, MAE 84.10). The smaller demographic groups—HISPANIC, ASIAN, OTHER—have lower errors than the two large groups (WHITE and Black), which most likely reflects narrower within-group LOS variance rather than a genuinely better model fit for those subgroups; we discuss this caveat in Section V.

*Comparison with prior work.* Wang et al. [3] report comparable relative AUC gaps across race and insurance in their classification setup; ours is the same phenomenon expressed in regression units. Lett et al. [11] demonstrate a 22.3% subgroup calibration-error reduction from intersectional debiasing on a similar ED prediction task, suggesting a concrete mitigation path our descriptive audit identifies but does not yet apply. The much smaller gender gap and the larger race/age gaps echo a well-documented pattern in the wider clinical-ML fairness literature [21], [22] that binary gender is a less common source of disparity than race or age in standard structured-data models.

## V. Discussion

### A. What the Results Mean

Three findings deserve to be foregrounded. First, the linear-vs-boosted ordering on this dataset is a counter-finding to the prevailing literature pattern that XGBoost and LightGBM dominate ED prediction. Read together with the SHAP result that one feature carries roughly half the total predictive signal, and with the very weak linear correlations of every other feature, the parsimonious explanation is that on a moderately sized dataset with one dominant linear driver and a weakly informative remainder, regularized linear regression has nothing to lose and boosting has too little interaction signal to gain. We expect the ordering to flip on much larger datasets or on datasets enriched with real-time queue-depth signals. Second, ESI stratification reveals that the single pooled metric overstates how well the model serves high-acuity patients and understates how well it serves low-acuity patients; reporting a single MAE without stratification is therefore quietly misleading for any clinical deployment. Third, the race/ethnicity MAE gap of 15.26 minutes is large enough to matter for patient communication—the difference between "tell me 90 minutes" and "tell me 105 minutes" is the difference between trust and distrust the next time the patient walks in.

### B. Strengths

The study has four strengths worth naming. (1) The dataset is fully public under a permissive license, the random seed is fixed, and the cleaned CSV plus analysis notebook are released with the paper, so the experiment is reproducible end-to-end. (2) The pipeline simultaneously delivers continuous regression, ESI stratification, SHAP, and a fairness audit—a combination not present in any of the twelve reviewed studies. (3) We include simple regularized-linear baselines and report honestly that they win, against the publication-bias gradient toward boosting in this literature. (4) Every demographic subgroup with at least 50 test-set records is reported.

### C. Limitations

We acknowledge five limitations. *Sample size.* 9,216 visits is modest compared with the 187,028 records of Wu et al. [9] or the ∼400,000 of MIMIC-IV-ED [4]; both the linear-vs-boosted ordering and the per-subgroup error estimates carry meaningful uncertainty at our sample size. *Single source.* The dataset comes from a single ED, so external validation on a different institution is required before the model can be claimed to generalize. *No queue signal.* The static feature vector cannot include real-time crowding indicators (current census, mean wait of patients ahead in queue, board status);

Wu et al. [9] report that exactly this feature is the strongest predictor in their model, and its absence in our setup likely caps the achievable $R^2$. *Limited explanatory power.* The pooled $R^2$ of 0.141 means 86% of the variance in LOS is unexplained by triage features alone, motivating future incorporation of NLP on chief-complaint text [6] and laboratory turnaround features. *Descriptive fairness only.* We report subgroup gaps but do not yet apply the intersectional-debiasing correction of Lett et al. [11]; that mitigation is left for follow-on work.

### D. Practical Implications

For ED operations leads, these results suggest that a Ridge model deployed at triage on this kind of feature set can deliver a patient-level estimate accurate to roughly $\pm 90$ minutes on average, with substantially better accuracy ($\pm 37$ min) for the lowest-acuity patients and substantially worse accuracy ($\pm 130$ min) for the second-highest acuity tier. Patient communication should respect this stratification: a confident "30 to 60 minutes" is realistic for ESI-5 patients but not for ESI-2 patients, who should be told something like "two to six hours, depending on workup." For equity researchers, the 15-minute race-related MAE gap is a concrete and reproducible target for a debiasing intervention. For clinicians, the SHAP ranking is a useful sanity check: the model is leaning on physiological severity, not on demographics or administrative features.

## VI. Conclusion and Future Work

We presented an end-to-end regression pipeline for predicting Emergency Department length of stay as a continuous variable, evaluated five models on 9,216 ED visits with an 80/20 holdout and 5-fold cross-validation, stratified results across all five ESI levels, applied SHAP TreeExplainer to interpret the best gradient-boosted model, and audited demographic fairness across race/ethnicity, gender, and age group. Ridge Regression produced the lowest test MAE (89.09 min, $R^2 = 0.141$), narrowly ahead of Random Forest and ahead of the two gradient-boosted models—a result that nuances the literature's general preference for XGBoost and LightGBM in this domain. SHAP confirmed that ESI acuity dominates the prediction signal. ESI stratification revealed an MAE gradient from 37 (ESI 5) to 130 (ESI 2) minutes that is invisible in pooled metrics. The fairness audit surfaced a 15.26-minute MAE gap across race/ethnicity and an 11.50-minute gap across age groups, with a near-zero gap across genders.

Three lines of follow-on work are most promising. First, incorporating NLP-processed chief-complaint text along the lines of Seo et al. [6] should add semantic signal that the static-feature formulation cannot capture. Second, replicating the experiment on MIMIC-IV-ED [4], [13] and on a second institutional dataset would test external validity. Third, applying the intersectional debiasing of Lett et al. [11] would translate the descriptive fairness gaps reported here into measurable mitigation outcomes. We also plan to layer rule extraction in the style of Sulaiman et al. [8] on top of SHAP to produce clinician-facing "if–then" explanations alongside the feature-attribution view.


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## APPENDIX

This appendix collects every numerical table referenced in the body of the paper. Each table is structured as a synthesis grouped by theme, method, or clinical role—rather than a flat list—so that the reader can scan for patterns across rows.

TABLE II
RAW DATA DICTIONARY, GROUPED BY CLINICAL ROLE ($N = 9{,}216$)

| Role | Field(s) | Type / Range |
|---|---|---|
| Identifier | `patient_id` | int / unique |
| Demographics | `age` | int / 0–100+ |
| | `gender` | cat. / M, F |
| | `race` | cat. / 5 levels |
| Arrival context | `arrival_hour` | int / 0–23 |
| | `arrival_transport` | cat. / 4 levels |
| Triage | `acuity` | ord. / 1–5 (ESI) |
| | `pain_score` | float / 0–10 |
| Vital signs | `heartrate` | float / bpm |
| | `sbp` | float / mmHg |
| | `dbp` | float / mmHg |
| | `temperature` | float / °F |
| | `o2sat` | float / % |
| | `resprate` | float / br/min |
| Disposition | `disposition` | cat. / 5 levels |
| | `was_admitted` | bin / 0/1 |
| Target | `ed_los_minutes` | float / minutes |

TABLE III
TARGET VARIABLE DISTRIBUTION, GROUPED BY DISTRIBUTION REGION

| Region | Statistic | Value (min) |
|---|---|---|
| Centrality | Mean | 189.8 |
| | Median | 153.7 |
| Spread | Std | 141.7 |
| Tails | Min | 8.6 |
| | 25th percentile | 95.6 |
| | 75th percentile | 241.6 |
| | Max | 2,644.9 |

TABLE IV
MISSING-VALUE PROFILE, GROUPED BY SEVERITY (TOTAL 5,156 CELLS, 3.3%)

| Severity | Field | Count | % |
|---|---|---|---|
| High (≥10%) | `temperature` | 1,105 | 11.99 |
| Moderate (8–10%) | `resprate` | 921 | 9.99 |
| | `o2sat` | 829 | 9.00 |
| | `pain_score` | 737 | 8.00 |
| Low (≤6%) | `dbp` | 552 | 5.99 |
| | `sbp` | 552 | 5.99 |
| | `heartrate` | 460 | 4.99 |

TABLE V
WRANGLING PIPELINE BEFORE VS. AFTER, GROUPED BY PROPERTY

| Property | Metric | Before | After |
|---|---|---|---|
| Volume | Rows | 9,216 | 9,216 |
| | Columns | 17 | 26 |
| Quality | Missing cells | 5,156 | 0 |
| | Missing % | 3.3 | 0.0 |
| Schema | Numeric features | 13 | 22 |
| | Categorical features | 4 | 5 |
| Target stability | Target mean (min) | 189.8 | 189.8 |
| | Target std (min) | 141.7 | 141.7 |
| | Target min (min) | 8.6 | 8.6 |
| | Target max (min) | 2,644.9 | 2,644.9 |

TABLE VII
ESI-STRATIFIED RIDGE PERFORMANCE, GROUPED BY ACUITY TIER

| Tier | ESI | $N$ | Mean LOS | MAE | RMSE | $R^2$ |
|---|---|---|---|---|---|---|
| High acuity | 1 | 179 | 315.7 | 113.33 | 152.26 | −0.158 |
| | 2 | 1,159 | 265.2 | 130.28 | 174.81 | −0.072 |
| Mid acuity | 3 | 4,184 | 215.9 | 101.96 | 138.38 | −0.006 |
| Low acuity | 4 | 2,706 | 145.0 | 69.96 | 101.62 | −0.015 |
| | 5 | 988 | 91.4 | 36.96 | 48.25 | −0.066 |

TABLE VIII
GLOBAL MEAN |SHAP| FOR LIGHTGBM, GROUPED BY FEATURE FAMILY

| Family | Feature | Mean \|SHAP\| |
|---|---|---|
| Triage | `acuity` | **46.07** |
| Vital signs | `o2sat` | 7.15 |
| | `dbp` | 6.27 |
| | `temperature` | 5.88 |
| | `resprate` | 5.20 |
| | `heartrate` | 4.96 |
| | `sbp` | 3.94 |
| Engineered physiology | `pulse_pressure` | 5.85 |
| | `shock_index` | 5.07 |
| Pain | `pain_score` | 3.70 |
| Demographics | `age` | 5.87 |
| | `race_encoded` | 3.49 |
| | `gender_encoded` | 1.40 |
| | `age_group_encoded` | 0.62 |
| Arrival / disposition | `arrival_hour` | 3.37 |
| | `arrival_transport_enc` | 2.15 |
| | `is_weekend` | 1.32 |
| | `was_admitted` | 1.14 |
| | `arrival_shift_encoded` | 0.96 |

TABLE I
SYNTHESIS OF THE TWELVE REVIEWED STUDIES, GROUPED BY THEME

| Theme | Study | Year / Method | Best Metric | Gap Left Open |
|---|---|---|---|---|
| Systematic reviews | Porto [1] | 2024 / BERT, SVM, LSTM | AUC ∼0.85+ | No regression; no SHAP |
| | Multi-author SLR [2] | 2025 / RF, XGBoost, LR | AUC 0.81–0.93 | Binary only; no public dataset |
| Empirical wait time / LOS | Wang (fairness) [3] | 2025 / XGBoost + SHAP | AUC ≈0.72 | Binary only; ESI-3 only |
| | Ricciardi [5] | 2024 / RF, DT, SVM | AUC ≈0.80 | Binary; no SHAP; no segmentation |
| | Seo [6] | 2024 / LightGBM + NLP | Best AUC-ROC overall | Single center; no fairness |
| | Wu [9] | 2025 / LightGBM (regression) | Lowest MAE / RMSE | Single center; no SHAP; no fairness |
| Benchmarks / interpretability | Xie [4] | 2022 / GB, RF, MLP | Beats MEWS/NEWS | No wait time; no SHAP; no fairness |
| | Sulaiman [8] | 2025 / GB + TE2Rules | Best classification | Binary; single center; no regression |
| Fairness / overcrowding / advanced AI | Vural [10] | 2025 / TSiTPlus, XCMPlus | MAE 4.19; $R^2$ 0.56 | Aggregate count; not patient-level |
| | Lett [11] | 2025 / XGBoost (intersect.) | AUROC 0.85 | Admission only; no regression |
| | Sun (ED-Copilot) [12] | 2024 / BioGPT + RL | Halved avg lab wait | Binary targets; no ESI segmentation |
| Interpretable (directly compared) | Wang (interp.) [7] | 2025 / XGB, RF, ANN, SVM | Best Acc/Recall/F1 | Binary; ESI-3 only; no public data |

TABLE VI
OVERALL MODEL PERFORMANCE, GROUPED BY MODEL FAMILY ($n_{\text{test}} = 1{,}844$; CV ON TRAINING FOLD)

| Family | Model | Test MAE | Test RMSE | Test $R^2$ | CV MAE | CV RMSE | CV $R^2$ |
|---|---|---|---|---|---|---|---|
| Linear / regularized | Linear Regression | 89.10 | 130.35 | 0.141 | 90.01 | 131.33 | 0.145 |
| | Ridge Regression | **89.09** | **130.34** | **0.141** | **90.01** | **131.33** | **0.145** |
| Bagged trees | Random Forest | 91.03 | 131.96 | 0.120 | 92.23 | 133.58 | 0.114 |
| Boosted trees | XGBoost | 93.80 | 134.91 | 0.080 | 94.81 | 137.18 | 0.065 |
| | LightGBM | 93.42 | 134.84 | 0.081 | 94.84 | 137.26 | 0.064 |
| Sanity floor | Mean baseline | 98.71 | — | — | — | — | — |

TABLE IX
FAIRNESS AUDIT BY RACE/ETHNICITY, GROUPED BY SUBGROUP SIZE

| Tier | Subgroup | $N$ | MAE | RMSE |
|---|---|---|---|---|
| Large ($N > 300$) | WHITE | 1,032 | 90.78 | 135.47 |
| | BLACK/AFR. AM. | 366 | 91.23 | 130.00 |
| Medium (100–300) | HISPANIC | 245 | 83.05 | 118.23 |
| | ASIAN | 133 | 88.01 | 124.12 |
| Small ($N < 100$) | OTHER | 68 | 75.97 | 103.64 |
| **Range (max−min)** | | | **15.26** | **31.83** |

TABLE X
FAIRNESS AUDIT BY GENDER (RIDGE, TEST SET)

| Subgroup | $N$ | MAE | RMSE |
|---|---|---|---|
| Female (F) | 981 | 88.61 | 137.02 |
| Male (M) | 863 | 89.64 | 122.31 |
| **Range (max−min)** | — | **1.03** | **14.71** |

TABLE XI
FAIRNESS AUDIT BY AGE GROUP, GROUPED BY LIFE STAGE

| **Stage** | **Subgroup** | $N$ | **MAE** | **RMSE** |
|---|---|---|---|---|
| Pediatric | $\leq$18 | 148 | 95.60 | 150.62 |
| Adult | 19–40 (Young) | 600 | 93.34 | 136.35 |
| | 41–65 (Middle) | 901 | 84.10 | 125.58 |
| Geriatric | $>$65 (Elderly) | 195 | 94.13 | 115.62 |
| **Range (max$-$min)** | | | **11.50** | **35.00** |